



mkat

**JOB TICKETING AND SITE SERVICE
MANAGEMENT APPLICATIONS**

Proposed Applications

Date of issue: 27 January 2025

Table of Contents

- Introduction4**
- Features of the Application4**
- Client Interface4**
- Admin Interface.....8**
- Technician App 10**
- mkat 15**

List of Figures

Figure 1 - Login	5
Figure 2 - Alternate Contact options	5
Figure 3 - Client Dashboard	6
Figure 4 - New Service Request	6
Figure 5 - Active Service Requests	6
Figure 6 - Client Reporting	7
Figure 7 - Client Equipment Management	7
Figure 8 - Client Services Menu	7
Figure 9 - Unprocessed Requests	8
Figure 10 - Ongoing Jobs	9
Figure 11 - Technician Login	11
Figure 12 - Technician Job Card	11
Figure 13 - Proof of Safety Compliance	11
Figure 14 - Documentation Completion	11
Figure 15 - Document Parts Installed	12
Figure 16 - Technician Job Completion	12
Figure 17 - Job Card Equipment Details	13
Figure 18 - Job Card Checks	13
Figure 19 - Job Card Readings	14

Introduction

This document describes MKat's proposed solution for managing job ticketing and service requests, an App called *Service Centre*. This solution allows **clients to log job requests, issue job cards, and oversee job completion**. Also described is the potential integration into a Petroleum Safety Training Application (PSTA) that allows individuals to undergo training to acquire safety badges that can be required to be assigned to a job card. The PSTA is not included in this document but will be referenced for how the solution may integrate and leverage its capabilities to expand its capabilities.

Features of the Application

The solution has three main user types, each of which will have a tailored experience in the application based on their roles. The following sections will group the proposal by the interface pages, which the specific users will interact with.

The three users are:

- Admin
- Technicians
- Clients

The following subsections will describe these users and the pages they interact with.

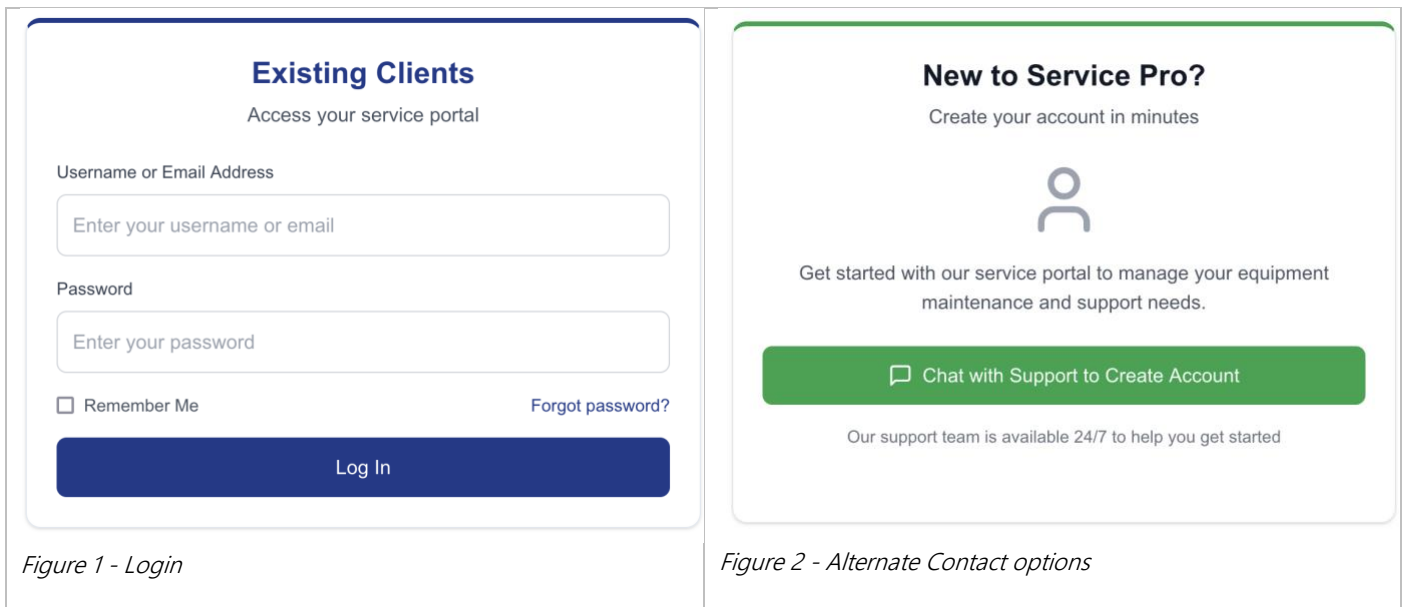
Client Interface

An app that allows the clients to log an enquiry for services to be rendered. This will be available from the website.

This is where any new job starts. When a client (Petrol Station) needs to request services, they have a range of contact options broken down into two categories: Automatic (Figure 1) and Manual (Figure 2). Both options are available from the website, the clients can either log in and request the service or be redirected to WhatsApp, where an automatic service will prompt the user for all the same details.

A phone number will be displayed on the website if the client chooses to use the manual option; this will involve them calling the admin staff, who will act on their behalf by capturing the request on the system. The admin interface will be discussed shortly.

A demonstration of the Client Interface can be found at: <http://102.37.23.105:3005/>



Once a user logs in (Figure 1), they will be greeted by their dashboard (Figure 3), where they can create a new service request. The creation of a new service request prompts the client for information pertaining to the job (Figure 4). These include:

- The location of the site
- The ID or serial number of the equipment that requires service (as well as visual evidence)
- A description of the issue
- The priority level required for the response.

Once the service request has been created, it will be added to the list of active service requests which the user can also view from their dashboard (Figure 5). A job can then be created and the admin team can create the job card and dispatch a maintenance team accordingly. This list will also depict the status and progress of the service requests as well as the technician who has been assigned to the job.

Jobs can be summarised by a progress report that the owners can download (including how many hours have been completed).

In addition to creating new service requests, clients can also review their history of completed jobs (Figure 6). The equipment on site can also be monitored (Figure 7).

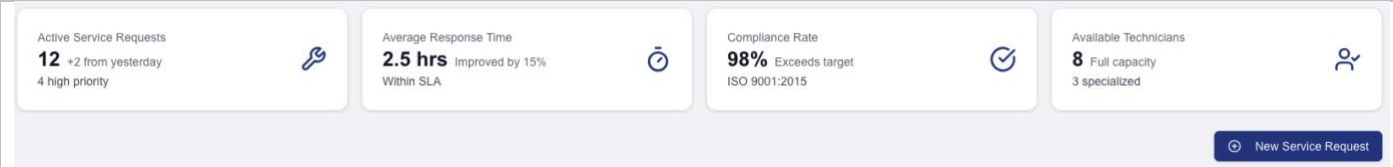


Figure 3 - Client Dashboard

Equipment Details

Site Location*

Equipment ID*

Select location

Enter ID or Serial Number

Response Priority

Critical

2hr Response

High

4hr Response

Medium

8hr Response

Low

24hr Response

Issue Details

Describe the issue, including: • Specific symptoms observed • Timeline of the issue • Related maintenance history • Environmental conditions

Visual Evidence

Add Photos

Figure 4 - New Service Request

Active Service Requests									
Real-time status of ongoing service requests									
Search by ID, site,...					All Statuses		All Priorities		
Reference	Site	Equipment	Issue	Priority	Status	Progress	Technician	ETA	Actions
SR-2024-001	Station Alpha	Fuel Dispenser	Flow Rate Deviation	High	In Progress	65%	John Smith	2 hours	
SR-2024-002	Beta Terminal	Storage Tank	Level Sensor Calibration	Medium	Assigned	20%	Sarah Johnson	4 hours	
SR-2024-003	Gamma Station	POS Terminal	Payment Processing Error	Critical	Overdue	0%	Unassigned	Overdue by 48 hours	
SR-2024-005	Epsilon Fuel Stop	Vapor Recovery	Compliance Check	High	Overdue	15%	David Chen	Overdue by 24 hours	
SR-2024-006	Zeta Highway Stop	Emergency Generator	Scheduled Maintenance	Medium	Scheduled	0%	Lisa Martinez	Next Week	
SR-2024-007	Eta Express	Price Sign	Display Malfunction	Medium	In Progress	45%	Tom Brown	3 hours	
SR-2024-008	Theta Services	Underground Piping	Pressure Test Required	Critical	On Hold	30%	James Wilson	Pending Parts	
SR-2024-009	Iota Junction	Car Wash Equipment	Water Pressure Low	High	Pending Customer	75%	Alex Thompson	Awaiting Approval	

Figure 5 - Active Service Requests

mkat - every bit counts

6

📄 Reports ^

🕒 Service History

📊 Analytics

📄 Compliance Docs

⚠️ Safety Records

Figure 6 - Client Reporting

🔄 Equipment Status ^

📁 Fuel Systems

⚙️ Pumps & Dispensers

📦 Tank Monitoring

📈 Performance

Figure 7 - Client Equipment Management

🔑 Service Center ^

📄 Active Tickets 5

🕒 Scheduled Service

🔧 Calibration

📄 Compliance

Figure 8 - Client Services Menu

Admin Interface

An app that allows the administrative staff to manage jobs on sites.

Once a service request is created by the user or by the admin staff on behalf of the client, it will appear on the Admin Interface until it is processed (Figure 9). The client requests will be handled manually by the admin staff, but an automation service and further interfaces can be provided. Once the admin staff have handled the requests, they use their interface to create job cards and assign them to technician teams (Figure 10).

A demonstration of this application can be found at: <http://102.37.23.105:3006/>

Unprocessed Requests

Ongoing Jobs

Search by ID, site...

Create New Request

Unprocessed Service Requests							
Service requests awaiting processing and assignment							
Reference	Site	Equipment	Issue	Priority	Technician	ETA	Actions
SR-2024-001	Sandton Depot	Fuel Dispenser	Flow Rate Deviation	High	Unassigned	Pending Processing	⋮
SR-2024-002	Durban Terminal	Storage Tank	Level Sensor Calibration	Medium	Unassigned	Pending Processing	⋮
SR-2024-003	Pretoria Station	POS Terminal	Payment Processing Error	Critical	Unassigned	Pending Processing	⋮

Figure 9 - Unprocessed Requests

Unprocessed Requests

Ongoing Jobs

Search by ID, site...

Create New Request

Ongoing Job Cards

Currently active service requests being processed

Reference	Site	Equipment	Issue	Priority	Technician	ETA	Actions
JC-2024-001	Cape Town Terminal	Vapor Recovery	Compliance Check	High	David Naidoo	2 hours	
JC-2024-002	Bloemfontein Stop	Emergency Generator	Scheduled Maintenance	Medium	Sarah van der Merwe	Next Week	

Figure 10 - Ongoing Jobs

As maintenance is performed, a detailed database will be built for each site, eventually establishing a record of every part on every site to allow easy search and glean which parts are problematic where. This approach will allow our algorithms to provide a preventative maintenance schedule for each pump on every site to say which parts need to be replaced when to minimise the chance of failure.

Technician App

An app that allows the technicians to log jobs and fulfil the required legislative and safety requirements that are involved in a Job

Once the admin team processes a service request and uses their interface to create and assign technicians to a job card, the technicians will be notified via SMS, email, WhatsApp and the Technician App.

This app will require the technicians to log in (Figure 11) and enter a job card number (Figure 12).

For a new job, all of the required documentation will be available for the technician to present to the client for signature (Figure 14).

Once the documentation has been completed, the app will prompt the technician for photographic evidence of the Personal Protective Equipment required to complete the job (Figure 13). Once all of these steps have been completed, the technician will be allowed to access additional menus that allow them to complete the job.

All parts which are installed or serviced have a brand and serial number which must be captured in the app along with photo evidence of both the new and old parts (Figure 15). Any maintenance or procedural documentation which cannot be digitised must be uploaded photographically as well. Furthermore, in the event of an accident, technicians will be able to compile and submit incident reports. These will help improve both training and the sites serviced.

The above can all be automated into a WhatsApp Service that automatically requests and captures the above information for the technician for ease of use.

Furthermore jobs that require more detailed forms can be assigned more detailed job cards as seen in Figure 17. All checks must be performed before work can commence (Figure 18) and the details of the job can be entered (Figure 19). These job cards are digitised for ease of use in the application and can be exported into a PDF format for documentation purposes. Should the document need to take a special format for legal reasons, a PDF viewer can host these forms in the application.

A demonstration of this application can be found at: <http://102.37.23.105:3002/>

Login

Enter your credentials to access your assigned jobs

Login

 Report Incident

Figure 11 - Technician Login

Job Card Entry

Enter or scan your job card number

Enter Manually

Scan QR Code

 Report Incident

Figure 12 - Technician Job Card

Safety Requirements

Complete safety documentation and PPE verification

PPE Verification

Safety Helmet

 Upload

Safety Boots

 Upload

Safety Gloves

 Upload

Confirm Safety Checks

 Report Incident

Figure 13 - Proof of Safety Compliance

Client Documentation

Obtain necessary client signatures and approvals

Required Signatures

Work Authorization Form

Capture Signature

Safety Compliance Form

Capture Signature

Continue to Work

 Report Incident

Figure 14 - Documentation Completion

Work Execution

Document parts and complete required procedures

🔑 Parts Documentation

Part Number

Serial Number

📷 Old Part

📷 New Part

📄 Additional Documentation

Work Notes

📷 Upload Additional Photos

Complete Job

🚨 Report Incident

Job Completion

Review and submit completed job

👍 Job Complete

All required documentation has been collected and verified.

Job Summary

Job Number:

Safety Checks: Completed

Client Signature: Obtained

Parts Logged: Verified

Save Draft

Submit Job

🚨 Report Incident

Figure 16 - Technician Job Completion

Figure 15 - Document Parts Installed

mkat - every bit counts

12

Equipment Details

Enter equipment and issue information

Equipment Information

Make & Model (e.g., Dresser Wayne Helix)

Serial Number

Grade (e.g., ulp93 and dsl50)

Contact Number

Issue Description

Continue to Safety Checks

 Report Incident

Figure 17 - Job Card Equipment Details

Safety Checklist

Complete all required safety checks before proceeding

Dealer's consent to
commence work?

Y

N

Protective clothing and safety
shoes are being worn?

Y

N

Barriers erected around the
work place?

Y

N

Fire extinguisher (9kg)
accessible within the work
place?

Y

N

Electrical supply isolated and
tagged?

Y

N

Fuel supply to the pump or
dispenser isolated?

Y

N

Continue to Work

 Report Incident

Figure 18 - Job Card Checks

Work Execution

Document work carried out and parts used

Work Details

Describe work carried out

Root Failure Cause

Not Pump Related

Meter Readings

Litre Tote Before

Litre Tote After

Parts Used

S/Code

Description

Add Another Part

Faulty Items Returned

S/Code

Description

Add Another Item

Complete Job

 Report Incident

Figure 19 - Job Card Readings

mkat

MKat specialises in data. We build the interfaces and databases that help you create, capture and store that information, not just so that's its safe, but so that you can leverage it. Data can reveal inefficiencies waiting to be optimised, trends waiting to be capitalised and insights that can be actioned.

If you are interested in any of our solutions and offerings, feel free to contact us.



Stephan Mulder

Director

stephanm@mkat.co.za

(+27) 83 397 0276